COMP6713 2026 – T1

This document will help you define the scope of your project. The following table will help you plan the scope of the project. You must accrue a minimum of 80 credits while completing the minimum credits in each of the four parts. Please upload the completed document along with the group details here: https://forms.office.com/r/c5R5BddBZF

PART A: Problem Definition **Minimum: 10 credits**			**Fill your project details here**
	Points	**Details**	
NLP Problem	5 credits per problem	Examples: Question-answering, Sentiment Analysis, etc. (Refer to course modules)	**1. Stance classification:** classifying financial text as hawkish, dovish, or neutral based on implied monetary policy position. **2. Sentiment classification:** classifying financial text as positive, negative, or neutral in terms of market sentiment
Text Source/Domain	5 credits per text source/domains	Examples: News Articles, Medical papers, social media posts, etc.	**1. Central bank communications:** official FOMC meeting minutes and statements from the U.S. Federal Reserve. **2. Financial news headlines:** short-form news text from financial media sources.
PART B: Dataset Selection **Minimum: 20 credits**			
Use two existing datasets	10 credits	Publicly available datasets.	**1. FOMC Hawkish-Dovish Dataset** (gtfintechlab, GitHub): FOMC texts labelled hawkish/dovish/neutral **2. Financial PhraseBank** (Takala, HuggingFace): financial news sentences labelled positive/neutral/negative
Create your own labelled dataset	20 credits	Correct selection of labels, inter-annotator agreement	
Use an existing lexicon	10 credits	Examples: WordNet, medical ontology	Loughran-McDonald Financial Sentiment Dictionary: word lists for positive, negative, and uncertainty categories, plus custom hawkish/dovish word

			lists for monetary policy stance. Used for rule-based classification and as additional features combined with TF-IDF.
PART C: Modelling			
Implement a rule-based or statistical model as a baseline	-	Essential. **Please keep this approach simple – this is only a baseline.**	**Baseline (rule-based/statistical):** TF-IDF + Logistic Regression applied to both tasks as a non-neural baseline.
Minimum: 30 credits			
Use an existing pre-trained/fine-tuned model	5 credits per model	You must compare the performance of multiple models to accrue credits, if you are only using available fine-tuned models or only prompting.	FinBERT, BERT-base-uncased, and RoBERTa-base evaluated zero-shot and few-shot on both datasets to assess the impact of domain-specific pre-training on financial NLP tasks.
Fine-tune a model based on a dataset	20 credits per out-of-the-box fine-tuning method	Examples: fine-tuned BERT, prefix-tuned LLaMA	FinBERT fine-tuned separately on each dataset using task-specific classification heads.
Extend a method	30 credits per extended fine-tuning method	Examples of extension: modification of the loss function, incorporation of structured ontology, prompting method other than zero/few-shot prompting, etc.	We extend the fine-tuning approach by training a unified multi-task model on both datasets simultaneously, with shared BERT encoder layers and separate task-specific heads. We additionally investigate weighted cross-entropy loss to address class imbalance in the FOMC dataset. We analyse whether joint training improves generalisation across domains compared to single-task fine-tuning.
Integrate a language model with external tools	20 credits	Usage of a library like LangChain/LangSmith	
PART D: Evaluation **Minimum: 20 credits**			
Quantitative Evaluation	10 credits	Appropriate metrics.	Per-class F1, macro-F1, accuracy, and confusion matrices for both tasks across all model variants

Qualitative Evaluation	5 credits	Examine mis-classified instances and produce common error types	Error analysis of misclassified examples, examining ambiguous sentences where hawkish/dovish signals co-occur, and cases of domain mismatch
Command line testing	5 credits	Interface to test out the system. This can be executed using an input argument or input file.	Command-line interface accepting raw text input and returning predicted label + confidence score
Demo	10 credits	A simple demo through Gradio (or equivalent).	Web interface where users can paste any financial sentence and receive stance + sentiment predictions with confidence scores

Deliverables

The project will be marked out of 25. The distribution is as follows.

1) Code-base: [10 marks]
    a. May be submitted as a Jupyter notebook(s) or a Python repository
    b. Must be well-commented and comprehensible to an NLP layperson.
    c. Assessors will attempt to manually run all components of the code.
2) Report [5 marks]
    a. Report must outline the choices made in parts A to D, as per the table above.
    b. Highlight results, inter-annotator agreement, qualitative analysis wherever applicable.
    c. Justify the choice of the model, tools and techniques, as applicable.
    d. The report must be comprehensible to an NLP layperson.
3) Presentation [5 marks]
    a. All teams must submit a presentation of their project.
    b. The presentation is intended for an advanced NLP audience.
    c. The presentation will be followed by Q&A by the assessor(s).
    d. Expected duration: (To be advised)
4) Individual Effort [5 marks]
    a. These marks will be separately awarded to every team member.
    b. The marks will be determined based on peer evaluation and Q&A with the course team.
    c. The course team might reach out to team members to discuss peer evaluation.